Machine Learning
Quiz Refresher: Chapter 5, Answer Sheet

Answers to the 20 recall questions, the 8 multiple-choice questions and the 5 exam-style problems

Machine Learning

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Answers

Round A

A1. Averaging / bagging (parallel randomised fits, average to cut variance); boosting (sequential corrections that reduce the current loss); stacking and mixtures (learn how to combine, globally or per input).

A2. Hard: each model votes its class, majority wins. Soft: average the posteriors, take the argmax. Soft voting needs calibration, because it adds probabilities: an overconfident model dominates the sum.

A3. $\mathrm{Var}(\bar{e}) = \sigma^2(\rho + (1-\rho)/M)$. $\rho = 0$: $\sigma^2/M \to 0$. $\rho = 1$: σ^2, no gain. In general a floor of $\rho\sigma^2$ remains.

A4. Different bootstrap samples, random feature subsets at each split, different model families, different random initialisations or hyperparameters, changing targets (residuals) across rounds.

A5. $T(x) = \sum_j \theta_j \mathbf{1}[x \in S_j]$: a partition into leaves with one constant each. Mean (squared), median (absolute), majority class (zero–one).

A6. $I(S) = 1 - \sum_c \hat{p}_c(S)^2$; 0 for a pure node; 0.5 for two classes at 50/50.

A7. $(j^*, t^*) = \arg\max_{j,t} [I(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R)]$. Greedy: it picks the best split now, never looking ahead to what later splits could achieve.

A8. Flexible (low bias) but high-variance, exactly what averaging fixes; fast to fit and easy to randomise; scale-free and handle mixed feature types.

A9. N draws with replacement from the N training examples. About 63.2% distinct: $\mathbb{P}(\text{omitted}) = (1 - 1/N)^N \to e^{-1} = 0.368$.

A10. $\hat{f}_{\mathrm{OOB}}(x^{(i)})$ averages the models whose bootstrap sample omitted i (about $0.368M$ of them); the OOB error averages the resulting losses. It is free internal validation, fine for tuning the forest, but once used to choose it is optimistic; a test set is still needed.

A11. At each split only m_{try} random candidate features are considered. $m_{\mathrm{try}} = d$ is plain bagging.

A12. Lower m_{try}: trees become weaker (each split has fewer good features to pick) but less correlated; the forest improves as long as ρ falls faster than σ^2 rises.

A13. $\mathrm{MDI}_j = \frac{1}{M} \sum_m \sum_{t:\mathrm{split}(t)=j} \Delta_t$, the impurity removed by splits on j, weighted by node size. Biased toward features with many split points, and computed on training data (rewards overfitting splits).

A14. I_j = error after shuffling feature j (on OOB or validation data) minus error before. Masking: if j and k are correlated, shuffling one leaves the other to carry the signal, so both look unimportant. Importance measures reliance of the fitted model, never causality.

A15. $F_m = F_{m-1} + \eta T_m$, $F_0 = \bar{y}$. Each tree is fitted to what the current sum gets wrong, so the trees are corrections to be added, not alternative predictions to be averaged.

A16. $r_m^{(i)} = -\partial \ell / \partial F(x^{(i)})$ at F_{m-1}: the direction that reduces the loss fastest at that example. For squared loss it is $y^{(i)} - F_{m-1}(x^{(i)})$, the ordinary residual.

A17. Rounds M, learning rate η, tree depth (or leaves). Roughly $\eta \propto 1/M$: halving η doubles the rounds needed. M by early stopping on validation loss, for each candidate (η, depth).

A18. The logit F, with $\hat{p} = \sigma(F)$. $g = \hat{p} - y$, $h = \hat{p}(1 - \hat{p})$.

A19. Newton leaf $w = \sum(-g)/(\sum h + \lambda)$; XGBoost $w_q^* = -G_q/(H_q + \lambda)$; gain $= \frac{1}{2}[G_L^2/(H_L + \lambda) + G_R^2/(H_R + \lambda) - G^2/(H + \lambda)] - \gamma$. λ penalises large leaf weights, γ penalises each additional leaf.

A20. Stacking: a meta-model with fixed weights β over base predictions, fitted on out-of-fold predictions (in-sample ones leak). Mixture of experts: a gate $g(x) = \text{softmax}(a(x))$ gives input-dependent weights, trained jointly with the experts on the mixture likelihood.

Round B

B1. **(b)** $1 \cdot (0.5 + 0.5/100) = 0.505$: the correlation floor 0.5 dominates.

B2. **(a)** $1 - (0.75^2 + 0.25^2) = 1 - 0.625 = 0.375$.

B3. **(c)** $1 - e^{-1} = 0.632$.

B4. **(d)** every split considers every feature, so the only randomness left is the bootstrap.

B5. **(c)** $F_1 = 5 + 0.1 \times 2 = 5.2$; the leaf's correction is shrunk by η.

B6. **(a)** $g = \hat{p} - y = 0.9$, $h = \hat{p}(1 - \hat{p}) = 0.09$: a confident wrong prediction has a large gradient.

B7. **(c)** $w^* = -G/(H + \lambda) = 6/4 = 1.5$ (without λ it would be 2).

B8. **(b)** in-sample predictions reward memorisation; the meta-model must see out-of-fold predictions so each base model is judged on data it did not train on.

Round C

C1. How much does averaging buy?

(a) Var $= 4(\rho + (1 - \rho)/10)$. Independent: 0.4. $\rho = 0.3$: $4(0.3 + 0.07) = 1.48$. $\rho = 0.9$: $4(0.9 + 0.01) = 3.64$.

(b) Limit $4\rho = 1.2$. $4(0.3 + 0.7/M) \leq 1.6 \iff 0.7/M \leq 0.1 \iff M \geq 7$.

(c) Hard: class 1 gets two votes, class 2 one $\Rightarrow$ class 1. Soft: mean $(0.417, 0.583) \Rightarrow$ class 2. Soft is more defensible: two lukewarm votes against one near-certain one, and averaging posteriors keeps that information. It rests on the three models being calibrated; an overconfident third model would hijack the vote.

(d) All models miss the same information (a feature nobody has, a label noise shared by all); the models are near copies (same family, same data, no randomisation); a distribution shift that moves every model's errors in the same direction.

C2. One split, two candidates

(a) Node: $1-(0.5^2+0.5^2)=0.5$. A left $(0.75, 0.25)$: 0.375; A right: 0.375. B left $(1,0)$: 0; B right $(1/3, 2/3)$: $1-(1/9+4/9)=4/9=0.444$.

(b) A: $\frac{20}{40}0.375+\frac{20}{40}0.375=0.375$, reduction 0.125. B: $\frac{10}{40}\cdot 0+\frac{30}{40}\cdot 0.444=0.333$, reduction 0.167. The greedy algorithm chooses B: a pure leaf is worth more than two balanced-but-mixed ones.

(c) $\Delta_t=\frac{N_t}{N}[I(S_t)-\ldots]=\frac{40}{200}\times 0.167=0.033$.

(d) Left: positive, $\hat{p}=1.0$ (from 10 examples). Right: negative, $\hat{p}(\text{pos})=1/3$.

(e) A feature with 500 values offers 499 thresholds, so on noisy data one of them will find a spurious reduction; a binary feature offers one. MDI is also computed on the training data that the tree overfits. Compute permutation importance on OOB or validation data instead: a spurious split loses nothing when the feature is shuffled.

C3. Bootstrap, out-of-bag and m_{try}

(a) $0.632\times 1{,}000\approx 632$ distinct examples. Each tree omits a given example with probability 0.368, so about $0.368\times 500=184$ trees are OOB for it.

(b) $0.632^{500}=e^{500\ln 0.632}=e^{-229}\approx 0$. Not a worry; with $M=10$ it would be $0.632^{10}=0.01$, one example in a hundred.

(c) No. The OOB error of the winner is the minimum of three noisy estimates, optimistic in the same way as a best-of-validation score. A test set used once is still the only unbiased estimate.

(d) $\mathbb{P}(\text{at least one informative})=1-\binom{395}{m}/\binom{400}{m}\approx 1-(1-5/400)^m$. $m=20$: $1-0.9875^{20}=1-0.778=0.22$. $m=100$: $1-0.9875^{100}=1-0.285=0.72$. With only 5 signal features out of 400, $m_{\text{try}}=\sqrt{400}=20$ makes four splits in five choose among pure noise: the trees are weak and the loss of strength outweighs the decorrelation. Expect 100 to win. The default assumes informative features are not that rare.

(e) A tree grown to purity has low bias and high variance; it memorises the noise in its own bootstrap sample. Inside a forest the memorised noise differs from tree to tree and averages out (that is what the variance formula buys), while the low bias is kept. Alone, the same tree would simply overfit.

C4. Two rounds of gradient boosting by hand

(a) $F_0=\bar{y}=24/4=6$. Residuals $r_1=y-6=(-4,-2,0,6)$.

(b) Left leaf $\{1,2\}$: mean -3; right leaf $\{3,4\}$: mean $+3$. $F_1=6+0.5\,T_1=(4.5,4.5,7.5,7.5)$. Residuals $r_2=(-2.5,-0.5,-1.5,4.5)$.

(c) Squared error of fitting r_2 with two leaf means:

split	leaf means	SSE	
$x\le 1$	$-2.5 \mid 0.833$	$0+20.7=20.7$	
$x\le 2$	$-1.5 \mid 1.5$	$2+18=20$	
$x\le 3$	$-1.5 \mid 4.5$	$2+0=2$	best

$T_2=-1.5$ for $x\le 3$, $+4.5$ for $x=4$. $F_2=F_1+0.5\,T_2=(3.75,3.75,6.75,9.75)$.

(d) F_0: $(16+4+0+36)/4 = 14$. F_1: $(6.25+0.25+2.25+20.25)/4 = 7.25$. F_2: $(3.06+0.06+0.56+5.06)/4 = 2.19$.

(e) $\eta = 1$: $F_1 = (3, 3, 9, 9)$, residuals $(-1, 1, -3, 3)$, MSE 5: a bigger step, lower training error after one round, but the whole leaf correction is committed at once. With $\eta = 0.5$ each round only takes half the step, so about twice the rounds are needed to reach the same fit: $\eta \propto 1/M$. Smaller steps let later trees revise earlier ones and usually generalise slightly better.

(f) Minimising $\sum_i \ell(y^{(i)}, F_{m-1}(x^{(i)}) + T(x^{(i)}))$ over a small step T is steepest descent in function space: the best direction at each example is $-\partial\ell/\partial F$. For squared loss $\ell = \frac{1}{2}(y-F)^2$ this is $y - F$. For absolute loss it is $\text{sign}(y - F)$: the tree fits the signs of the residuals, not their size, which is why absolute-loss boosting is robust to outliers.

C5. An XGBoost leaf and whether to split it

(a) $g = \hat{p} - y$: $-0.7, -0.4, +0.2$, so $G = -0.9$. $h = \hat{p}(1-\hat{p})$: $0.21, 0.24, 0.16$, so $H = 0.61$.

(b) $w^* = -G/(H+\lambda)$: $\lambda = 0$: $0.9/0.61 = 1.475$; $\lambda = 1$: $0.9/1.61 = 0.559$. The step is positive (two of three examples are positives being under-predicted). Loss removed: $\frac{1}{2}G^2/(H+\lambda) = 0.5 \times 0.81/1.61 = 0.252$.

(c) $G_L = -1.1$, $H_L = 0.45$; $G_R = 0.2$, $H_R = 0.16$. Gain $= \frac{1}{2}[1.21/1.45 + 0.04/1.16 - 0.81/1.61] - \gamma = \frac{1}{2}[0.834 + 0.034 - 0.503] - \gamma = 0.183 - \gamma$. $\gamma = 0.1$: $0.083 > 0$, split. $\gamma = 0.2$: $-0.017 < 0$, keep the leaf. γ is the price of a leaf.

(d) With the first-order approximation $\ell(F + w) \approx \ell(F) + gw$ the objective is linear in w: it has no minimum, only a direction, so the step size needs a separate line search per leaf. Adding $\frac{1}{2}hw^2$ makes it a quadratic bowl, minimised in closed form at $w = -G/(H+\lambda)$, and the same quadratic gives the gain formula for choosing the split.

(e) Lower η alone: the model now needs more rounds, so at fixed M it underfits. Shallower trees alone: the validation minimum moves to a different M, which was not re-tuned. Fewer rounds alone: may stop before the validation minimum for the current η and depth. Right procedure: search over (η, depth) and choose M for each by early stopping on validation loss, then compare configurations at their chosen M.